Day 4 afternoon — Hierarchical models

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Friday 13 November 2026 · Kroghstræde 3, room 1.104 · group rooms 4.129 and 5.149

Grouped data · complete, no and partial pooling · exchangeability · shrinkage · the funnel and non-centring · predicting for a known or a new person

Today

08:45	**Lecture A**	Two unknowns: joint, marginal, conditional · one new attempt
09:45	**Exercise A**	Normal model in Stan · joint draws · average vs attempt · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Grouped data · pooling · shrinkage · hierarchical models in Stan
13:30	**Exercise B**	Three pooling models · shrinkage · known vs new person · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from the morning plenary

Open points from Peer review A

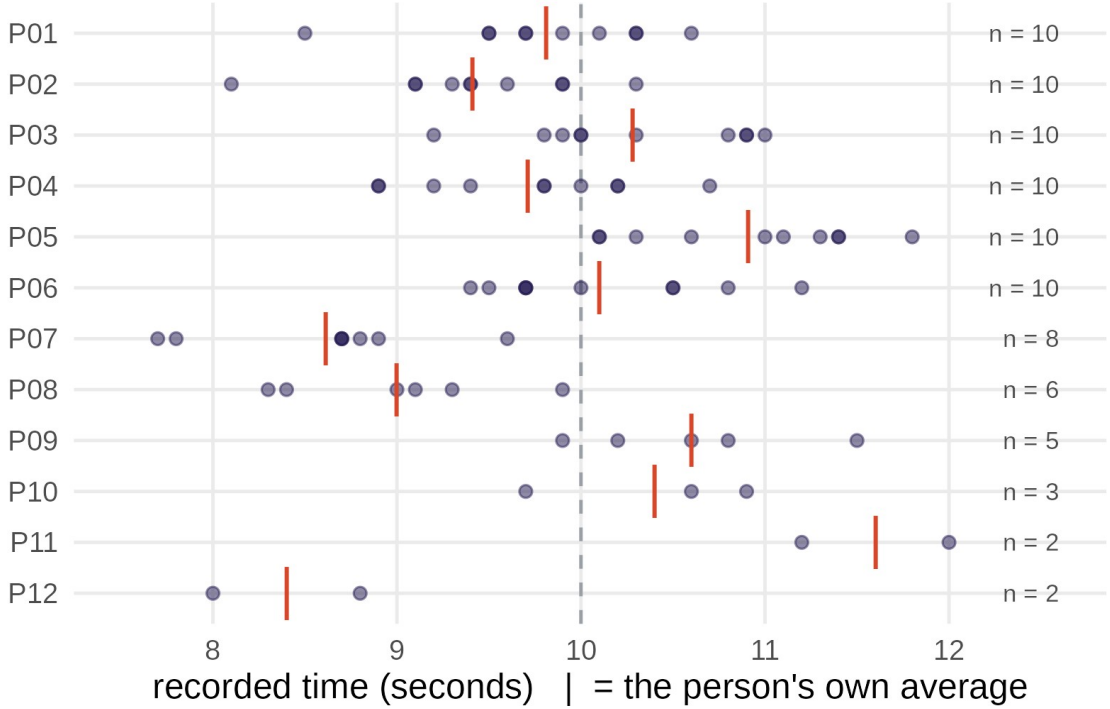
Fill in during lunch: the one or two questions the plenary did not settle.

1. …
2. …

By the end of this lecture you can

1. **Recognise** grouped data: attempts within people
2. **Compare** complete, no and partial pooling
3. **Explain** shrinkage as a compromise between a person's data and the group
4. **Fit** a hierarchical model in Stan, and fix a funnel by non-centring
5. **Predict** for a person we know and for a new person

Now the whole class



- 12 people, 2 to 10 attempts each (illustrative numbers).
- Some are fast, some slow: people differ.
- Each person also varies from attempt to attempt.

Questions

What is P11's average, from 2 attempts?

What will a new person do?

Attempts are grouped within people

y_{ij} = attempt i of person j

Between people

Each person has her own average θ_j.

The θ_j differ: some people count fast.

Within a person

Attempts scatter around θ_j with SD σ.

Same σ for everybody (a simplification).

- Two attempts by the same person are more alike than attempts by two people: the 92 attempts are not 92 independent draws from one distribution.
- But given θ_j, a person's attempts are independent: $y_{ij} \mid \theta_j, \sigma \sim \text{Normal}(\theta_j, \sigma)$. (conditional independence)

Three ways to use the class

Complete pooling

$y_{ij} \sim \text{Normal}(\mu, \sigma)$

One average for everybody.
Ignores that people differ.
P11 gets the class average.

No pooling

$y_{ij} \sim \text{Normal}(\theta_j, \sigma)$
$\theta_j \sim \text{Normal}(10, 2)$

A separate average per person, each with a fixed prior.
Learns nothing from the others.
P11 rests on 2 attempts.

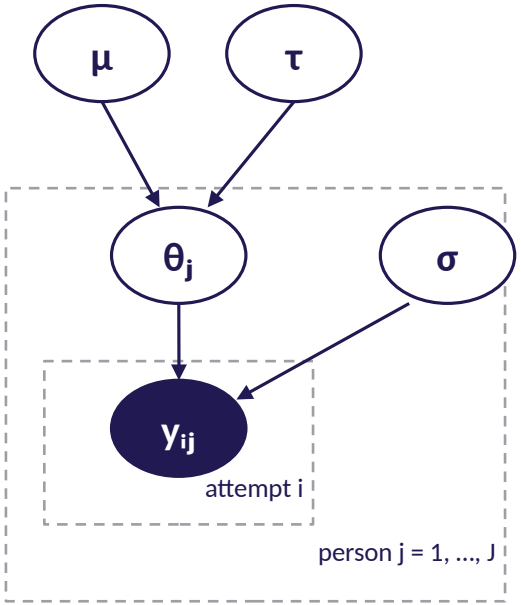
Partial pooling

$y_{ij} \sim \text{Normal}(\theta_j, \sigma)$
$\theta_j \sim \text{Normal}(\mu, \tau)$

Separate averages from a common distribution; μ and τ are learned.
The others tell us which averages are plausible.

Partial pooling sits between the two: it lets the data decide how alike people are (τ).

The hierarchical model



level	model	meaning
attempts	$y_{ij} \sim \text{Normal}(\theta_j, \sigma)$	within a person: σ
people	$\theta_j \sim \text{Normal}(\mu, \tau)$	between people: τ
priors	$\mu \sim \text{Normal}(10, 2)$ $\tau \sim \text{Normal}^+(0, 1)$ $\sigma \sim \text{Normal}^+(0, 2)$	μ: average of the averages

"Hierarchical": the prior for θ_j has its own unknown parameters μ and τ, learned from all people.

Why one common distribution? Exchangeability

The assumption

Before seeing the data, nothing tells us that P03 is faster than P07: swapping the labels would not change our beliefs.

Then we may treat $\theta_1, \ldots, \theta_J$ as draws from one distribution.

When it fails

We know something that separates people: half had feedback, some are musicians.

→ Put it in the model as a predictor (Days 5–6).

A new person

Predictions for a new person assume she is like these people.

A PhD class is not every population.

Two sources of information about θ_j

Suppose μ, τ and σ were known. What do we learn about θ_j, person j's average?

Her own attempts

Her average $\bar{y}_j$ of n_j attempts:

$\bar{y}_j \mid \theta_j \sim \text{Normal}(\theta_j, \sigma/\sqrt{n_j})$

(an average of n values has SD $\sigma/\sqrt{n}$, Day 3)

precision $a = n_j / \sigma^2$

The other people

The population distribution:

$\theta_j \sim \text{Normal}(\mu, \tau)$

It acts as her prior.

precision $b = 1 / \tau^2$

Precision = 1 / variance: how sharp a piece of information is. Many attempts (large n_j) or little attempt-to-attempt spread (small σ) → large a. People very alike (small τ) → large b.

The estimate for person j combines a likelihood and a prior — exactly as on Day 2.

Where the weight w comes from

1. Posterior ∝ likelihood × prior. On the log scale:

   $$\log p(\theta_j \mid y) = -a\,(\bar{y}_j - \theta_j)^2 / 2 - b\,(\theta_j - \mu)^2 / 2 + \text{constant}$$

2. This is a parabola in θ_j, so the posterior is normal and its mean is the peak.

3. At the peak the slope is zero: $a\,(\bar{y}_j - \theta_j) + b\,(\mu - \theta_j) = 0$

4. Solve for θ_j: $\theta_j = (a\,\bar{y}_j + b\,\mu) / (a + b) = w_j\,\bar{y}_j + (1 - w_j)\,\mu$, $w_j = a / (a + b)$

$$w_j = (n_j / \sigma^2) / (n_j / \sigma^2 + 1 / \tau^2)$$: her data's share of the total precision.

Shrinkage: a compromise, again

$$E[\theta_j \mid y] \approx w_j \cdot \bar{y}_j + (1 - w_j) \cdot \mu$$

- The estimate lies between her own average and μ: it is pulled ("shrunk") towards μ.
- $1 - w_j$ is the fraction of the way to μ.
- Few attempts → small a → small w_j → pulled more.
- Same idea as Day 2's "The posterior mean is a compromise"; now the prior is learned from the other people.

attempts n_j	w_j	moves
10	0.96	4%
3	0.87	13%
2	0.82	18%

σ = 0.61, τ = 0.91, μ = 9.91 (posterior means)

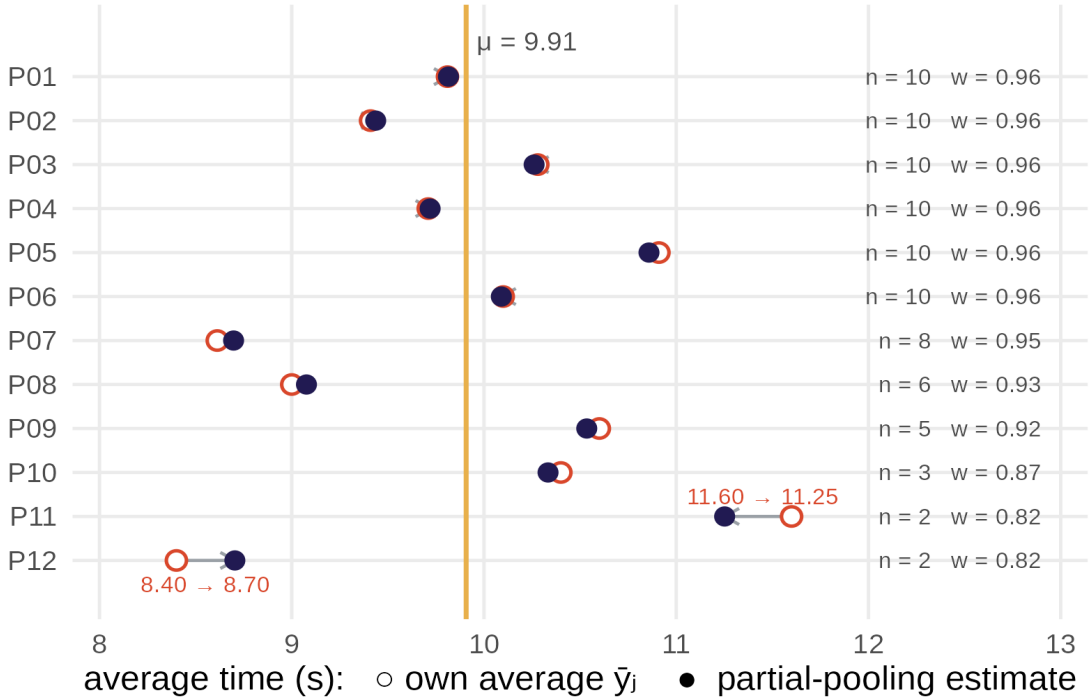
P11: 2 attempts, $\bar{y}$ = 11.60

0.82 × 11.60 + 0.18 × 9.91 = 11.29

(the full Stan fit: 11.25)

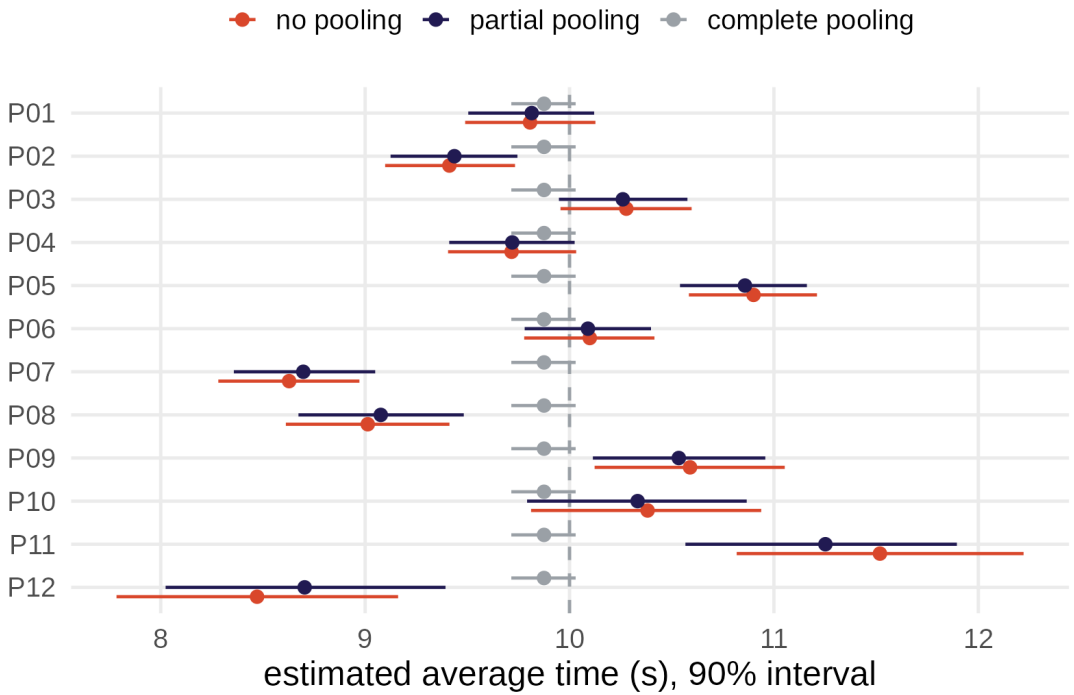
Few attempts → small weight on her own average → pulled more towards μ.

Shrinkage in the class



- Each arrow: from her own average $\bar{y}_j$ (○) to the estimate (●), towards μ (yellow line).
- Arrow length = $(1 - w_j) \times$ distance from $\bar{y}_j$ to μ.
- 10 attempts, w = 0.96: hardly any move.
- P11 and P12, 2 attempts, w = 0.82, and far from μ: the longest arrows.
- P10 also has few attempts but is close to μ: short arrow.

Three models side by side



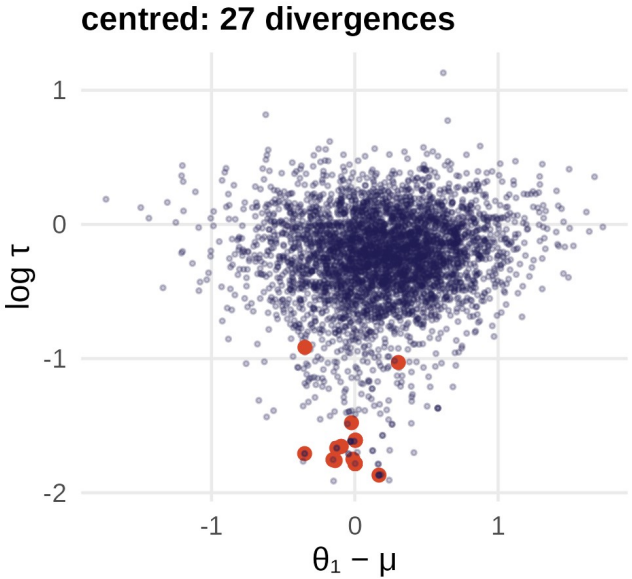
- Complete pooling: one value for all; misses P05 and P07.
- No pooling: wide intervals for P11 and P12.
- Partial pooling: close to no pooling where there are many attempts, narrower where there are few.

The hierarchical model in Stan

```
data {
  int<lower=1> N;              // all attempts
  int<lower=1> J;              // people
  array[N] int<lower=1, upper=J> person;
  vector[N] y;
}
parameters {
  real mu;
  real<lower=0> tau;
  vector[J] theta;             // person averages
  real<lower=0> sigma;
}
model {
  mu ~ normal(10, 2);
  tau ~ normal(0, 1);
  sigma ~ normal(0, 2);
  theta ~ normal(mu, tau);           // population
  y ~ normal(theta[person], sigma);  // likelihood
}
```

- person: for each attempt, the number of the person who made it.
- theta[person]: a vector of length N, each attempt's own person average.
- Two lines carry the hierarchy: the population and the likelihood.

A warning sign: the funnel

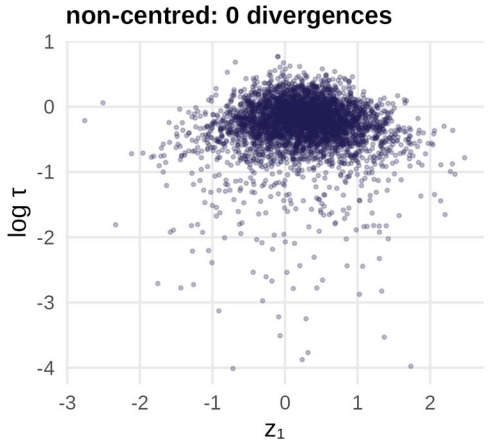


- Same model, but each person with only 2 attempts.
- When τ is small, all θ_j must lie close to μ: a narrow neck. When τ is large, they may spread: a wide mouth.
- One step size cannot suit both: 27 divergences (red), all in the neck (Day 3).

Divergences in hierarchical models usually mean a funnel, not a bug.

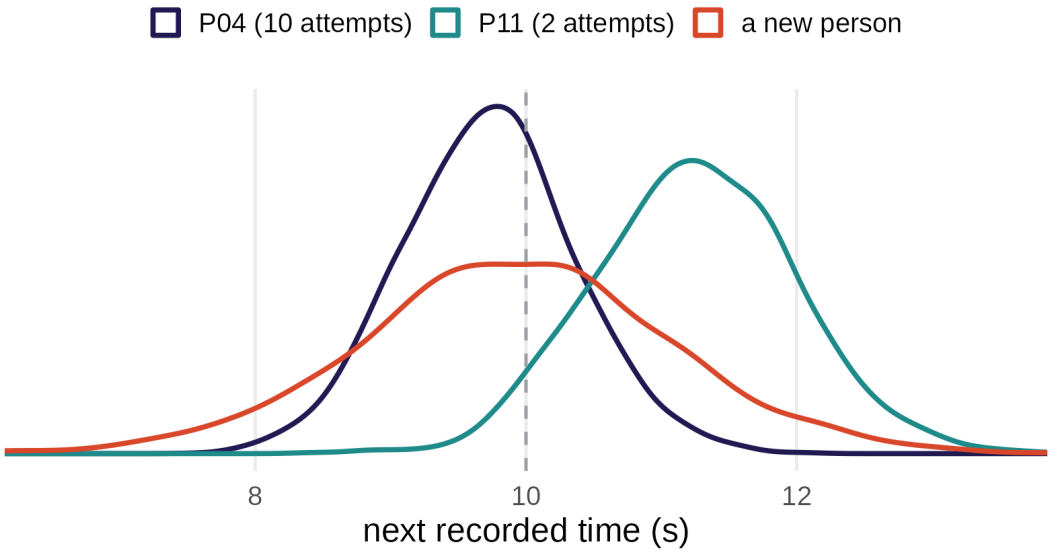
Non-centred: same model, easier shape

```
parameters {
  real mu;
  real<lower=0> tau;
  vector[J] z;
  real<lower=0> sigma;
}
transformed parameters {
  vector[J] theta = mu + tau * z;
}
model {
  ...
  z ~ std_normal();
  y ~ normal(theta[person], sigma);
}
```



- $\theta_j = \mu + \tau z_j$, $z_j \sim \text{Normal}(0, 1)$: still $\theta_j \sim \text{Normal}(\mu, \tau)$.
- Stan samples z: its shape does not depend on τ.
- 2 attempts each: 27 → 0 divergences.
  - With all attempts both versions work.

Predicting: a person we know, or a new person



next attempt by	90% interval
P04 (10 attempts)	8.7 – 10.8
P11 (2 attempts)	10.0 – 12.5
a new person	8.0 – 11.8

Known person j:
$\tilde{y} \sim \text{Normal}(\theta_j, \sigma)$: centred on her own, shrunk average
New person:
$\theta_new \sim \text{Normal}(\mu, \tau)$, then $\tilde{y} \sim \text{Normal}(\theta_new, \sigma)$: centred on μ

A new person adds the between-person spread τ to the within-person spread σ.

Worked example: shrinkage and prediction

Question 1

P10 made 3 attempts, own average 10.40 s.

With $\mu = 9.91$, $\sigma = 0.61$ and $\tau = 0.91$: compute w and the shrunken average.

Question 2

Next attempt, 90% intervals: P04 8.7–10.8, P11 10.0–12.5, new person 8.0–11.8.

(a) Why is the new person's interval the widest?

(b) Why is P11's centred near 11.25, not at $\mu = 9.91$ and not at her own 11.60?

Question 3

Which of the three models would you use to predict a new person, and why not the other two?

Worked example: answers

Question 1

$n/\sigma^2 = 3 / 0.61^2 = 8.06$; $1/\tau^2 = 1 / 0.91^2 = 1.21$

$w = 8.06 / (8.06 + 1.21) = 0.87$

$0.87 \times 10.40 + 0.13 \times 9.91 = 10.34$ (fit: 10.3)

Question 2

(a) For P04 only the attempt spread σ remains: width ≈ $2 \times 1.64 \times \sigma = 2.0$ s. For a new person her average is unknown too, so τ adds: $2 \times 1.64 \times \sqrt{(\tau^2 + \sigma^2)} = 3.6$ s.

(b) A known person's prediction starts from her own, shrunk average: 11.60 pulled towards 9.91 gives 11.25. With only 2 attempts that average is less certain, so her interval is also wider than P04's.

Question 3

Partial pooling: its population distribution Normal(μ, τ), learned from all people, describes a new person. Complete pooling lumps the two spreads into one σ and treats all 92 attempts as independent, so it is over-confident about μ (and predicts P04 like a stranger). No pooling has no distribution for a new person at all.

Exercise B · 13:30 – 15:00

- Fit complete, no and partial pooling to the class's attempts.
- Check the diagnostics of all three fits before comparing.
- Shrinkage plot: who moves towards the group, and why?
- Predict the next attempt for a person in the data and for a new person.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

15:00

peer review B

If Stan does not compile on your laptop: continue on UCloud or a group member's laptop. New roles this block.

Peer review B · 15:00 – 15:45

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

Points that need longer go to Monday's follow-up.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

Daily diary · 15:45 – 16:00

menti.com · code: ____ ____

1. Three things I have learned today
2. Three things I am not yet very familiar with